



Help you make and receive calls

Primitive mobile communication technology consisted of CB radios and walkie-talkies. A CB radio system consists of central antenna with around 20-30 channels and the phone transmits data on a particular channel to the tower which then retransmits it and can be received by any one logged on to that channel. A walkie-talkie is also used for communication but it has a single channel and no tower to retransmit its signal, hence it has a limited range. Both walkie-talkies and CB radios are half-duplex devices. That is, two people communicating on a CB radio use the same frequency, so only one person can talk at a time. That's why you always hear soldiers saying "over" after they finish their sentence so that the other person can start talking. Also limited number of conversations can take place simultaneously because of the limited number of channels.

On the other hand, a cell phone is a full-duplex device. That means that you use one frequency for talking and a second, separate frequency for listening. Both people on the call can talk at the same time. The cellular system divides the city into small cells (25 sq km) so thousands of people can use their cell phones simultaneously. A cell phone also has considerable large number of channels to communicate. Each carrier in each city also runs one